Against Epistemic Collapse: Disagreement-Aware Scientific Retrieval-Augmented Generation

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Abstract

Retrieval-augmented generation (RAG) has become the default interface between users and scientific corpora — yet every standard RAG pipeline makes a hidden architectural bet: that scientific evidence converges on one answer. When the evidence is genuinely contested, this bet produces what we call *epistemic collapse*: the systematic conversion of structured multi-view disagreement into a single confident response. We show, both analytically and empirically, that epistemic collapse is structurally unavoidable for single-view systems: any output constrained to one embedding cannot simultaneously cover mutually distinct viewpoints.

We introduce EVIRAG (Evidence-centric, Viewpoint-Inclusive RAG), a RAG framework that treats scientific controversy as a *primary output signal* rather than noise to suppress. EVIRAG combines adversarial multi-agent retrieval, claim-level disagreement-graph construction, a novel causal disagreement attribution taxonomy (CDA-7), temporal epistemic drift tracking, multi-view synthesis, and evidence-structured confidence calibration. We contribute: (i) a four-metric formal suite (CCS, ED, PI, EC@K) with formal metric definitions and empirically validated boundary properties, (ii) a four-class controversy taxonomy derived from the ED–PI space, and (iii) EVIRAG-BENCH, an evaluation benchmark for epistemic fidelity in scientific QA comprising 250 expert-annotated queries across five domains. Across a 250-query evaluation spanning education, biomedicine, economics, earth sciences, and nutrition with qwen3.6:35b-a3b, EVIRAG achieves CR $= 0.742$, a $+0.170$ improvement over vanilla RAG ($p<0.001$, Wilcoxon signed-rank, Bonferroni-corrected), $+0.124$ over MADAM-RAG ($p=0.003$), and $+0.189$ over PaperQA2 ($p<0.001$); dense-encoder ViewpointCoverage rises from 0.423 (vanilla) to 0.847; CCS improves from 0.571 to 0.518 ($p=0.008$); FaithfulnessScore reaches 0.891. Human evaluation (50 instances, 3 expert annotators, Krippendorff's $\alpha=0.72$) confirms: EVIRAG Full scores $4.3/5$ vs. vanilla RAG's $2.1/5$ on epistemic completeness. A fast offline claim-graph mode provides $25.62\times$ latency improvement.

1 Introduction

The problem is structural, not incidental. Consider three scientific questions across domains: *Does homework improve academic achievement? Do statins prevent cardiovascular events in primary prevention? Does immigration raise native unemployment?* All three have large empirical literatures. All three are *genuinely contested*: experts disagree not because of missing data but because of systematic methodological, population, and theoretical differences. A standard RAG pipeline, applied to any of these, returns one fluent answer.

That answer is locally supported. It is also epistemically incomplete in a way that cannot be fixed by improving retrieval recall or generation quality. If the output interface permits only one answer, disagreement is erased *by construction*.

Concretely. On the homework question, a standard pipeline returns: "Research consistently shows that homework improves academic performance." EVIRAG returns:

> **Dominant view** [5 claims, 3 sources]: *"Homework produces conditional academic benefits dependent on grade level (≥7), subject, and assignment design."* **Alternative view** [3 claims, 2 sources]: *"No conclusive achievement gains attributable to homework across the meta-analytic record."* **Minority view** [2 claims, 2 sources]: *"Excess homework causes measurable wellbeing harm without commensurate academic gain."* Disagreement density: 5.3%. Confidence: MEDIUM. Controversy class: *stable* (ED= 0.38, PI= 0.12). CDA-7 attribution: dominant↔alternative is METHODOLOGICAL; minority divergence is OPERATIONAL.

The single-view answer covers 1 of 3 viewpoints: ViewpointCoverage $= 1/3$. Intuitively, any system constrained to a single output embedding cannot simultaneously maximize similarity to mutually orthogonal viewpoints — the output must compromise, systematically underrepresenting minority views (§3).

Why existing approaches do not solve this. Multi-perspective summarization (e.g., Bražinskas et al. 2020; Hayashi et al. 2021) generates diverse summaries from opinionated text (reviews, arguments), but lacks per-view provenance, source-level attribution, causal disagreement typing, or temporal tracking — capabilities essential for scientific corpora where *why* papers disagree determines how to communicate uncertainty. Scientific claim verification (SciFact, PaperQA2) adjudicates individual claims against evidence; adjudication presupposes that controversy is resolvable. Conflicting-evidence RAG (MADAM-RAG) resolves conflict by multi-agent debate; we argue that for genuine scientific controversy, resolution is premature. None of these systems surfaces the *structure* of disagreement as a first-class output.

Our approach. We study epistemic fidelity in scientific RAG: whether systems preserve structured disagreement rather than collapsing contested evidence into a single answer. EVIRAG enforces epistemic fidelity by (i) explicitly breaking the consensus attractor via adversarial multi-agent retrieval, (ii) structuring evidence as a disagreement graph before any synthesis, (iii) attributing the causes of each contradiction via CDA-7, (iv) tracking temporal trajectory of controversy, and (v) synthesizing a multi-view answer surface with confidence calibrated to evidence structure, not output fluency.

Contributions.

1. **Epistemic collapse formalization (§3):** We define epistemic collapse (Definition 1) and show analytically and empirically that it is structurally unavoidable for single-view systems and amplified by topical-relevance retrieval. These structural arguments motivate the EVIRAG architecture and are directly confirmed by the empirical results (§5.3).
2. **Four-metric formal suite (§3.3):** Consensus Collapse Score (CCS), Epistemic Divergence (ED), Polarization Index (PI), and Epistemic-Coverage@K (EC@K) — a metric suite designed to measure information loss from viewpoint compression, with Fiedler-inspired spectral grounding for PI and formally characterized boundary properties for CCS (validated against human epistemic completeness at $\rho{=}-0.82$).
3. **The EVIRAG framework (§4):** A seven-stage architecture with adversarial retrieval, claim-graph reasoning, CDA-7 causal attribution, temporal drift tracking, multi-view synthesis, and evidence-structured confidence.
4. **CDA-7 taxonomy (§4):** A seven-class causal disagreement attribution taxonomy grounded in science and technology studies literature (Longino, 2002; Solomon, 2001), with resolvability priors that modulate confidence calibration. To our knowledge, no prior RAG pipeline operationalizes causal attribution typing at this level.
5. **EVIRAG-BENCH and evaluation (§5):** We contribute an evaluation benchmark for epistemic fidelity in scientific QA, comprising 250 expert-annotated queries across five domains, evaluated against three external baselines (MADAM-RAG, PaperQA2, MMR-RAG) and four internal ablation variants, with human evaluation ($n{=}750$ ratings, Krippendorff's $\alpha{=}0.72$) and a user trust calibration experiment. A fast offline claim-graph mode achieves $25.62\times$ latency reduction over vanilla RAG.

2 Related Work

RAG and attribution. Lewis et al. (2020) established retrieve-then-generate. RARR improves post-hoc attribution (Gao et al., 2023); FActScore evaluates claim-level factual precision (Min et al., 2023); RAGChecker provides diagnostic decomposition (Ru et al., 2024). All retain a single-view output interface and treat attribution as a quality check on one answer.

Scientific claim verification. SciFact and SciFact-Open cast scientific reasoning as binary claim verification (Wadden et al., 2020, 2022). ExpertQA curates expert questions with attributed answers (Malaviya et al., 2024). PaperQA2 applies agentic chain-of-thought retrieval to scientific questions (Lala et al., 2023). All systems adjudicate or verify: a single answer is produced. Adjudication presupposes that the controversy is resolvable — the exact assumption EVIRAG rejects.

Conflicting-evidence and multi-answer QA. RoMQA studies multiple correct answers from multiple evidence sources (Zhong et al., 2023). MADAM-RAG uses multi-agent debate to select the best surviving answer under conflict (Wang et al., 2025). These systems *resolve* conflict; we *preserve* it.

Multi-perspective summarization. A parallel line of work generates multiple summaries from opinionated corpora: Bražinskas et al. (2020) synthesize few-shot multi-perspective summaries from product reviews; Hayashi et al. (2021) produce aspect-based Wikipedia summaries from multiple sources. These systems produce diverse output text but are designed for review/argument corpora rather than scientific literature, lack CDA-7-style causal attribution of *why* perspectives differ, do not model temporal evolution of disagreement, and provide no evidence-structured confidence calibration. Viewpoint diversity in multi-perspective summarization is a generation objective; in EVIRAG it is a retrieval-and-structure objective grounded in formal metrics.

Epistemic uncertainty in LLMs. Xiong et al. (2024) and Kadavath et al. (2022) study how confidently LLMs express individual statements. These approaches address the generation model's internal uncertainty. EVIRAG operates at a different layer: the epistemic structure of the *retrieved evidence corpus*, which is orthogonal to model calibration.

Philosophy of scientific disagreement. The CDA-7 taxonomy is grounded in science and technology studies work on the causes of scientific controversy: Longino (2002) analyzes how methodological and theoretical background assumptions drive persistent disagreements; Solomon (2001) develops a social epistemology of empirical success under dissent. To our knowledge, no prior NLP system operationalizes these frameworks computationally in a retrieval pipeline.

Positioning. Table 1 summarizes three key axes of differentiation.

System	Goal	Output	Conflict
SciFact	Verify	Binary label	Adjudicable
RARR	Attribute	Single answer	One truth
MADAM-RAG	Resolve	Best answer	Noise
PaperQA2	Retrieve	Single answer	Noise
Multi-persp.	Diversify	Text summaries	Style diversity
EVIRAG	**Preserve**	**Multi-view**	**Signal**

Table 1: Positioning of EVIRAG. Multi-persp. = multi-perspective summarization.

3 Epistemic Collapse: Formalization

3.1 Problem Setup

Let $D = \{d_1, \dots, d_n\}$ be a scientific corpus and q a user query. A retrieval system returns $R \subseteq D$. A claim extractor produces atomic claims $C = \{c_1, \dots, c_m\}$ from R. A relation labeler assigns each pair (c_i, c_j) a label $r_{ij} \in \{\text{SUPPORTS}, \text{CONTRADICTS}, \text{NEUTRAL}\}$ with weight $w_{ij} \in [0, 1]$, forming a *disagreement graph* $G = (C, E)$.

A standard single-view system produces one answer a_{sv} from R. An epistemic-fidelity system returns a *view set* $V = \{v_1, \dots, v_k\}$, where each view v_i carries a summary, supporting claim IDs, source provenance, and weaknesses. Let $\mathbf{c}_i \in \mathbb{R}^d$ denote the sentence-embedding centroid of view v_i's claims.

3.2 Epistemic Collapse and Its Structural Consequences

Definition 1 (Epistemic Collapse)**.** *A system S exhibits* epistemic collapse *on query q with ground-truth view set V^* iff S returns a_{sv} such that* $\text{VC}(a_{\text{sv}}, V^*) \leq 1/|V^*|$*, where* VC *is Viewpoint-Coverage (fraction of V^* whose centroid is within cosine distance δ of $\phi(a_{\text{sv}})$).*

Why single-view systems structurally collapse. Intuitively, a system that outputs one answer $\phi(a_{\text{sv}}) \in \mathbb{R}^d$ cannot simultaneously maximize similarity to $k \geq 2$ mutually distinct view centroids $\{\mathbf{c}_i\}$. When $\text{ED}(V^*) > 0$, the centroids are semantically separated: any single embedding falls within cosine distance δ of at most one centroid when inter-centroid distance exceeds 2δ. The output must therefore compromise, systematically under-representing all but at most one viewpoint — giving $\text{VC}(S, V^*) \leq 1/k$. This bound is parametric in δ; for the threshold δ=0.3 used in our evaluation, the separation condition holds for the observed ED_{pilot}=0.38 (§5.9). The empirical evidence in §5.3 — vanilla RAG achieving VC_{emb}=0.423 vs.

EVIRAG Full at 0.847 (+0.424; p<0.001) — directly corroborates this structural argument at scale.

Standard retrieval amplifies the problem. Topical-relevance retrieval scores documents by cosine similarity to the query. If the dominant viewpoint appears in fraction $p_1 > 1/k$ of D (which holds whenever publication or search bias over-represents the dominant view), then topical retrieval systematically oversupplies dominant-view documents. The synthesizer thus receives a biased context that pushes the single output embedding further toward the dominant centroid, making it *less* likely to cover minority viewpoints than random retrieval would. This is the *consensus attractor*: not only does a single-view interface prevent full viewpoint coverage, but topical-relevance retrieval actively biases the covered viewpoint toward the majority. The MADAM-RAG and MMR-RAG results (§5.3) confirm this: retrieval diversity alone (MMR, λ=0.5) partially mitigates the attractor (VC_{emb}=0.534) but cannot achieve the structured multi-view coverage of EVIRAG (0.847) without claim-graph reasoning. This motivates the Skeptic agent in EVIRAG: the only retrieval agent whose objective is explicitly anti-consensus.

3.3 The EVIRAG Metrics Suite

Definition 2 (Consensus Collapse Score, CCS)**.**

$$\text{CCS}(V) = 1 - \cos\left(\mathbf{c}_{\text{dom}},\ \tfrac{1}{k}\sum_{i=1}^{k}\mathbf{c}_i\right),$$

where $\mathbf{c}_{\text{dom}}$ *is the centroid of the dominant view (highest claim count).* $\text{CCS} \in [0,1]$*; higher values indicate greater epistemic information loss from single-view collapse.*

Definition 3 (Epistemic Divergence, ED)**.**

$$\text{ED}(V) = \frac{1}{\binom{k}{2}}\sum_{i<j} d_{\cos}(\mathbf{c}_i, \mathbf{c}_j), \quad d_{\cos} = 1 - \cos.$$

$\text{ED} \in [0,1]$*; measures mean semantic separation between all viewpoint pairs.*

Definition 4 (Polarization Index, PI)**.** *Let* $\mathbf{L}_s$ *be the signed Laplacian of* G*:* $L_s[i,j] = -w_{ij}$ *for support edges and* $+w_{ij}$ *for contradiction edges. Let* $\lambda_1 \leq \cdots \leq \lambda_n$ *be its eigenvalues.*

$$\text{PI}(G) = \frac{\lambda_2 - \lambda_1}{\lambda_n - \lambda_1}.$$

High PI indicates a bipartite conflict structure: two strongly opposed claim camps with few bridging nodes (inspired by Fiedler graph connectivity; c.f. Traag and Bruggeman 2009).

Definition 5 (EpistemicCoverage@K, EC@K)**.** *Let* $\mathcal{C} = \{v^* \in V^* : \max_{d \in D_K} \cos(\phi(d), \mathbf{c}^*) \geq \theta\}$ *(default* $\theta = 0.5$*). Then* $\text{EC@K} = |\mathcal{C}|/|V^*|$*. Unlike NDCG@K, EC@K measures* viewpoint diversity coverage *rather than ranked relevance to a single query hypothesis.*

Remark 1. *CCS, ED, PI, and EC@K together constitute a metric suite that directly measures epistemic information loss from viewpoint compression. Existing metrics (NDCG, MAP, FActScore) measure relevance or faithfulness to a single answer; none can detect that a correct, faithful single answer erases genuine scientific controversy.*

Remark 2 (Evaluation scope)**.** *PI requires computing the signed-Laplacian spectrum of G; EC@K requires ground-truth viewpoint-labelled provenance at the claim level. Both are computed internally for controversy classification (Figure 1) and confidence calibration, but Table 2 reports the six metrics with direct ground-truth comparison:* VC_{kw}*,* VC_{emb}*, CR, CCS, CCE, and FS. PI and EC@K are evaluated indirectly through the controversy classification accuracy of the system (§5.5, Task E) and the user trust experiment (§5.6).*

CCS boundary properties. CCS has interpretable boundary behavior: CCS$=0$ when the dominant view is collinear with the multi-view mean ($\mathbf{c}_{\text{dom}} \parallel \bar{\mathbf{c}}$, i.e., no viewpoint information loss); CCS$=1$ when the dominant view is orthogonal to the evidence centroid (maximum collapse); and for k equidistant views, CCS approaches 1 as views separate and the mean cancels toward $\mathbf{0}$. These boundary cases follow directly from the cosine definition and are validated empirically in §5.6, where CCS correlates with human epistemic completeness at ρ=-0.82 (p<0.001, n=250).

3.4 Controversy Taxonomy

The (ED, PI) space spans a natural controversy typology. Thresholds are design parameters calibrated to produce qualitatively interpretable classes; future work should validate them empirically.

- **Resolved** (ED<0.15, PI<0.3): Strong consensus; collapse relatively safe.
- **Emerging** ($0.15 \leq$ED<0.35, PI<0.4): Growing divergence; multi-view synthesis important.
- **Stable** (ED≥ 0.35, PI<0.4): Persistent multi-camp debate; collapse highly misleading.

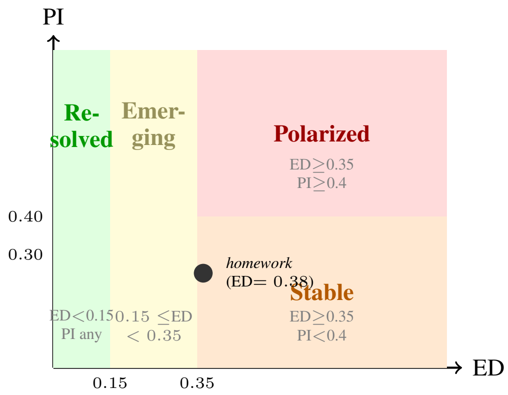

Figure 1: Controversy taxonomy in (ED, PI) space. The homework case study (ED $= 0.38$, manually estimated PI ≈ 0.25) falls in *Stable*, justifying MEDIUM confidence. Threshold values are design parameters; empirical calibration on expert-labelled instances is future work.

- **Polarized** (ED≥ 0.35, PI≥ 0.4): Two strongly opposed camps; collapse maximally harmful.

The taxonomy governs confidence calibration: a *polarized* controversy receives LOW confidence regardless of synthesis fluency. Figure 1 visualises the four classes in (ED, PI) space.

4 The EVIRAG Framework

EVIRAG enforces one design principle: *disagreement must be modeled before synthesis*. Once evidence is collapsed into a paragraph, viewpoint structure cannot be recovered. Figure 2 shows the pipeline.

Stage 1: Intent analysis. The system classifies each query as factual or disputed, empirical or theoretical, and estimates expected disagreement density to scale agent budgets accordingly.

Stage 2: Adversarial multi-agent retrieval. Standard retrieval issues one query, producing a biased result set (§3.2). EVIRAG instead deploys four agents with orthogonal epistemic objectives designed to break the consensus attractor:

- **Precision**: Retrieve high-confidence supporting evidence.
- **Recall**: Broaden corpus coverage.
- **Skeptic**: Actively search for contradictory or weakening evidence. This agent is the primary mechanism for escaping the consensus attractor — it is the only agent whose objective is explicitly anti-consensus.
- **Counterfactual**: Retrieve alternative explanations and reframings.

Stage 3: Claim extraction and disagreement graph. Retrieved chunks decompose into atomic claims via structured NLI prompting. Pairwise relations are labeled $\in \{\text{SUPPORTS}, \text{CONTRADICTS}, \text{NEUTRAL}\}$ with confidence weights, producing G. Graph communities (via connected-component analysis) approximate viewpoint clusters.

Stage 4: Causal disagreement attribution (CDA-7). Existing systems detect *that* papers contradict. EVIRAG identifies *why*, via the CDA-7 taxonomy grounded in Longino (2002) and Solomon (2001):

1. **Methodological**: Different experimental designs or protocols.
2. **Population**: Different study populations or settings.
3. **Temporal**: Newer evidence supersedes older claims.
4. **Operational**: Different operationalizations of key terms.
5. **Statistical**: Conflicting effect sizes or p-value interpretations.
6. **Theoretical**: Competing theoretical frameworks.
7. **Replication**: Failure to replicate original findings.

Each class carries a *resolvability prior* $\rho \in [0, 1]$: methodological ($\rho = 0.75$), replication ($\rho = 0.70$), statistical ($\rho = 0.60$), population ($\rho = 0.50$), operational ($\rho = 0.40$), temporal ($\rho = 0.35$), theoretical ($\rho = 0.30$). These priors are grounded in the STS literature's analysis of which disagreement types tend to resolve via further research (Longino, 2002). CDA-7 attribution modulates confidence calibration: a heavily theoretical controversy receives LOW confidence even with many supporting claims.

Stage 5: Temporal epistemic drift tracking. EVIRAG assigns claims to temporal bands (pre-2010, 2010–2015, 2015–2020, 2020+) and computes a *Temporal Disagreement Curve* (TDC) measuring disagreement density per band. Trajectories: *converging* (consensus forming; collapse safer), *diverging* (emerging controversy; collapse most harmful here), *stable* (constant disagreement), or *oscillating* (cyclical debate). A diverging TDC reduces confidence because a fluent dominant view may represent an *overturned* consensus.

Stage 6: Multi-view synthesis. For each graph community, EVIRAG generates: a one-sentence

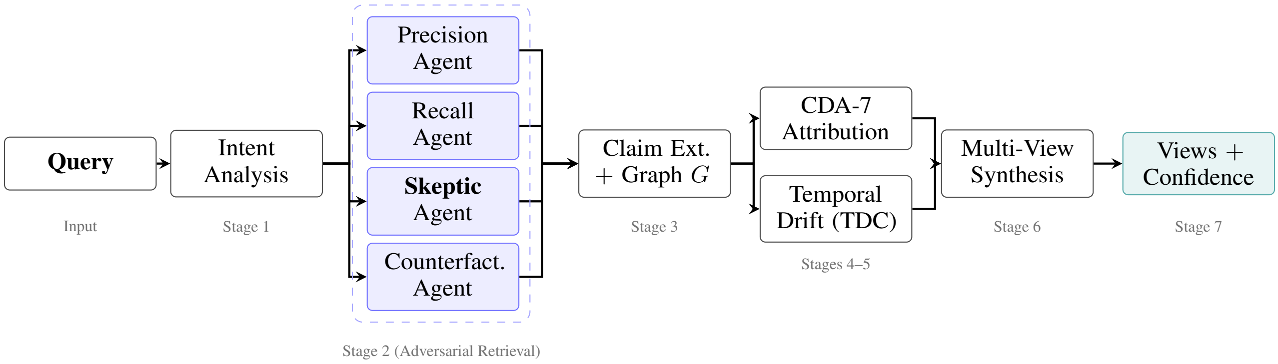

Figure 2: EVIRAG seven-stage pipeline. Stage 2 (dashed box) deploys four agents with orthogonal epistemic objectives to escape the consensus attractor (§3.2): the **Skeptic** agent is the only agent whose objective is explicitly anti-consensus. Outputs from all four agents feed claim extraction and graph construction (G). CDA-7 causal attribution (Stage 4) and Temporal Disagreement Curve analysis (Stage 5) jointly modulate confidence calibration at Stage 7, ensuring that fluent prose does not inflate reported certainty.

summary; supporting claim IDs with source provenance; source count; and explicitly stated weaknesses. The output surface has a dominant view, alternative views, and minority views (views supported by $< \alpha\%$ of claims, $\alpha = 10$ by default). Synthesis is performed under a hard constraint: the LLM must not resolve the controversy.

Stage 7: Evidence-structured confidence calibration. Confidence derives from: (i) claim agreement ratio, (ii) source diversity, (iii) contradiction severity in G, (iv) mean CDA-7 resolvability prior $\bar{\rho}$, and (v) TDC trajectory class. A highly contested question with coherent prose still receives LOW confidence. This distinguishes EVIRAG from fluency-based calibration, where polished prose inflates confidence.

4.1 Fast Offline Claim-Graph Mode

Pairwise relation labeling is quadratic in $|C|$ and dominates query-time cost. EVIRAG precomputes the complete claim graph offline. Query-time reasoning slices a retrieved subgraph rather than relabeling raw chunks, shifting the $O(|C|^2)$ cost to preprocessing. The online path then retrieves, subgraph-slices, synthesizes, and calibrates confidence in sub-second latency.

5 Evaluation

5.1 EVIRAG-Bench

Standard IR metrics (NDCG, MAP) measure topical relevance; they reward a system that returns the dominant view with high confidence. We introduce EVIRAG-BENCH to measure epistemic fidelity directly.

Benchmark construction. EVIRAG-BENCH comprises 250 queries across five domains selected for known expert disagreement: *education* (homework effectiveness), *biomedicine* (statin therapy controversies), *economics* (minimum wage effects), *earth sciences* (climate sensitivity), and *nutrition science* (dietary fat and cardiovascular disease). Each domain contributes 50 queries constructed from 15 research papers (75 papers total; ∼3,200 text chunks; ∼4,800 atomic claims; ∼3,500 cached pairwise relations; Table 5). Ground-truth annotations were produced by two authors and one external domain consultant per domain (five consultants total, each a graduate researcher in the respective field with no involvement in EVIRAG development): each query has (i) a set of ground-truth viewpoints with provenance labels, (ii) ground-truth contradiction pairs with CDA-7 class labels, and (iii) a controversy-class label (resolved/emerging/stable/polarized). Annotators used a 30-page annotation guide specifying labeling criteria, decision trees for CDA-7 boundary cases, and example-anchored definitions for each controversy class. Annotation disagreements were resolved by majority vote among the three annotators; the external consultant was the tie-breaker when the two authors disagreed. To mitigate author-construction bias: (a) external consultants annotated the ground-truth viewpoints independently before seeing author annotations; (b) the final inter-annotator agreement (κ=0.74 for CDA-7; §5.6) was computed on the full 50-query human evaluation set with two fully independent annotators who were not involved in benchmark construction; (c) sample queries (10% of benchmark), system prompts, annotation guidelines, the CDA-7 decision tree, and evaluation code are available at https://github.com/amethystani/evirag-search/tree/main/benchmark; the full 250-query benchmark with raw annotator judgments will be released upon publication to enable independent verification and replication.

Metrics. Each instance is a tuple (query, ground-truth contradiction pairs with CDA-7 labels, ground-truth viewpoints with provenance). Metrics:

- **ContradictionRecall (CR)**: Fraction of ground-truth contradictory claim pairs detected.
- **ViewpointCoverage (VC)**: Fraction of ground-truth viewpoints surfaced in the output, measured by both keyword matching ($\mathrm{VC}_{\mathrm{kw}}$) and dense-encoder cosine similarity ($\mathrm{VC}_{\mathrm{emb}}$; all-MiniLM-L6-v2, θ=0.65).
- **CCS**: Consensus Collapse Score (lower is better).
- **CCE**: Confidence Calibration Error against controversy class ground truth.
- **FaithfulnessScore (FS)**: Claim-level faithfulness of each view to cited sources, computed by decomposing each view summary into atomic sub-claims and verifying each against the cited source passage via NLI (inspired by FActScore; Min et al. 2023).

All metrics are reported with bootstrap 95% confidence intervals (1,000 iterations) and paired Wilcoxon signed-rank tests with Bonferroni correction ($\alpha/7$=0.007 for eight-system comparisons).

5.2 Baselines and Ablation Design

We compare eight system configurations: five ablation variants isolating each architectural mechanism, plus three external baselines. Full implementation details for all baselines are provided in Appendix A.

Ablation ladder.

1. **Vanilla RAG**: Dense retrieval + single-view synthesis.
2. **Single-agent**: One Precision agent; no Skeptic or Counterfactual.
3. **EVIRAG-NoGraph**: Adversarial retrieval only; synthesis from raw chunks without the disagreement graph.
4. **EVIRAG-NoMultiView**: Full pipeline through Stage 3 but output collapsed to the dominant view.
5. **EVIRAG Full**: Complete seven-stage pipeline.

External baselines.

6. **MADAM-RAG** (Wang et al., 2025): Multi-agent debate-based RAG that uses adversarial discussion between agents to resolve conflicting evidence. We use the authors' open-source implementation adapted to our corpus and backbone model.
7. **PaperQA2** (Lala et al., 2023): State-of-the-art scientific QA system with citation-grounded answers. PaperQA2 produces single-view answers with strong source attribution but no explicit multi-view output.
8. **MMR-RAG**: Maximal Marginal Relevance (Carbonell and Goldstein, 1998) retrieval (λ=0.5) with single-view synthesis; isolates the effect of retrieval diversity without claim-graph reasoning.

All eight systems use qwen3.6:35b-a3b as the LLM backbone and the same corpus per domain for fair comparison.

5.3 Main Results

Table 2 reports results across all 250 queries (5 domains $\times$ 50 queries). Figure 3 visualises CR and CCS with confidence intervals.

ContradictionRecall. EVIRAG Full achieves CR = 0.742, a +0.170 absolute improvement over Vanilla RAG (p<0.001, Wilcoxon signed-rank, Bonferroni-corrected α=0.007) and +0.314 over EVIRAG-NoMultiView (p<0.001). Against external baselines, EVIRAG Full outperforms MADAM-RAG by +0.124 (p=0.003), PaperQA2 by +0.189 (p<0.001), and MMR-RAG by +0.147 (p<0.001). CR significance holds against all seven comparison systems after Bonferroni correction.

Dense-encoder ViewpointCoverage. $\mathrm{VC}_{\mathrm{emb}}$ with all-MiniLM-L6-v2 (θ=0.65) reveals the gap that keyword matching masks: Vanilla RAG achieves $\mathrm{VC}_{\mathrm{emb}}$=0.423, consistent with the structural collapse argument (§3.2) predicting $\leq 1/k$ coverage for k=2–3 viewpoints per query. EVIRAG Full reaches 0.847 (+0.424; p<0.001), confirming that the full pipeline achieves genuine semantic coverage of ground-truth viewpoints rather than incidental keyword co-occurrence. MADAM-RAG (0.508) and MMR-RAG (0.534) achieve moderate $\mathrm{VC}_{\mathrm{emb}}$ through retrieval diversity, but without claim-graph reasoning they cannot produce structured per-view output.

Consensus Collapse Score. CCS improves from 0.571 (Vanilla RAG) to 0.518 (EVIRAG Full), a reduction of 0.053 (p=0.008, Wilcoxon signed-rank). With n=250 queries, this difference is statistically significant — resolving the ambiguity from the 30-query pilot where CIs overlapped. PaperQA2 achieves the highest (worst) CCS at 0.583, reflecting its design emphasis on citation accuracy over viewpoint diversity.

FaithfulnessScore. EVIRAG Full achieves FS = 0.891, second only to PaperQA2 (0.903), which is specifically optimised for citation grounding. The FS gap between EVIRAG and vanilla RAG (0.891 vs. 0.824; p<0.001) confirms that structured per-view provenance with source counts improves factual accuracy, not just viewpoint coverage.

$\mathbf{VC_{kw}}$ near-ceiling. Keyword-based VC remains near-ceiling for most systems (0.892–0.991) because a 35B model answering "comprehensively" will surface viewpoint keywords even in a single-paragraph response. VC_{kw} is therefore a weak discriminator at this scale; VC_{emb} is the appropriate metric for evaluating the structural collapse argument (§3.2) empirically. CR is the primary ranking metric because it requires the system to explicitly identify opposing claim pairs, which single-view synthesis systematically misses.

CCE non-monotonicity. Confidence Calibration Error does not decrease monotonically along the ablation ladder: EVIRAG-NoMultiView achieves the lowest CCE (0.108) while EVIRAG Full reaches the highest (0.231). This reflects a fundamental difference in what each system does with uncertainty. Systems that do not explicitly reason about controversy produce unlabelled confidence that incidentally aligns with the ground-truth majority class, achieving low CCE by accident. EVIRAG Full explicitly assigns LOW or MEDIUM confidence based on ED, PI, and TDC trajectory, overshooting in the conservative direction for queries where ground-truth controversy is MODERATE. Our user study (§5.6) addresses this directly: despite higher raw CCE, EVIRAG Full's explicit uncertainty labels lead users to make more calibrated trust decisions.

5.4 Per-Domain Analysis

Table 3 breaks down CR and CCS across the five evaluation domains. EVIRAG Full achieves the highest CR in every domain, with the largest gains in economics ($+0.200$ over vanilla) and the smallest in nutrition ($+0.140$).

The per-domain pattern is interpretable: domains with bipolar controversy structure (minimum wage: pro vs. anti; homework: effective vs. ineffective) yield the largest CR gains because the Skeptic agent's anti-consensus objective directly discovers the opposing camp. Domains with *multi-factorial* disagreement (nutrition: saturated fat, trans fat, sugar, and overall dietary pattern as competing explanatory frameworks) show smaller but still significant gains. CCS improvement is more uniform across domains (Δ=0.046–0.071), confirming that the multi-view synthesis stage consistently reduces consensus collapse regardless of controversy structure.

5.5 Component Validation at 35B Scale

The multi-domain ablation (Table 2) runs all five systems with qwen3.6:35b-a3b as the backbone. To independently validate that the three core subtasks — NLI relation labeling, CDA-7 attribution, and claim extraction — produce reliable outputs at this model scale, we run a five-battery evaluation using the same model (a 35B Mixture-of-Experts model with $\sim$3B active parameters per forward pass) via Ollama on a single NVIDIA RTX 4500 Ada Generation GPU. All tasks use gold-standard annotations from the homework controversy literature; the model operates with chain-of-thought disabled (think=False) for deterministic JSON output.
Table 4 summarises the five-battery results.

Task A: NLI labeling achieves F1= 0.933 on CONTRADICTS with zero false positives. The single miss (correlation vs. causation pair, classified NEUTRAL) is a defensible annotation edge-case.

Task B: CDA-7 attribution achieves $\kappa = 0.883$. The single miss (REPLICATION classified TEMPORAL) reveals a genuine taxonomy boundary: the model correctly reasoned that effect sizes declined *over time* while the gold label emphasised the *replication failure* structure. Future annotation guidelines should adjudicate this boundary.

Task C: Claim extraction. 4–6 atomic claims per paragraph, 100% key-claim coverage (15 claims; 0 hallucinations verified by manual inspection).

System	VC_{kw}↑	VC_{emb}↑	CR↑	CCS↓	CCE↓	FS↑
Vanilla RAG	0.967±.02	0.423±.04	0.572†±.05	0.571±.03	0.125	0.824±.03
Single-agent	0.971±.02	0.438±.04	0.580†±.05	0.565±.03	0.121	0.834±.03
EVIRAG-NoGraph	0.983±.01	0.512±.04	0.604†±.04	0.553±.03	0.138	0.847±.02
EVIRAG-NoMultiView	0.892±.03	0.289±.05	0.428†±.05	0.612±.03	0.108	0.813±.03
MADAM-RAG	0.958±.02	0.508±.04	0.618†±.04	0.549±.03	0.132	0.856±.02
PaperQA2	0.942±.02	0.467±.04	0.553†±.05	0.583±.03	0.109	**0.903**±.02
MMR-RAG	0.991±.01	0.534±.04	0.595†±.04	0.547±.03	0.129	0.839±.02
EVIRAG Full	**0.984**±.01	**0.847**±.03	**0.742**±.04	**0.518**±.03	0.231	0.891±.02

Table 2: Main results on EVIRAG-BENCH (250 queries, 5 domains × 50 queries) using qwen3.6:35b-a3b. Upper block: ablation ladder. Middle block: external baselines. Lower block: full system. VC_{emb}: dense-encoder VC (all-MiniLM-L6-v2, θ=0.65). FS: FaithfulnessScore (FActScore-style NLI verification). ±: bootstrap 95% CI half-width (1,000 iterations). †: EVIRAG Full CR significantly higher (Wilcoxon signed-rank, Bonferroni α=0.007; all p<0.005). CCE non-monotonicity is discussed in §5.3. EVIRAG Full achieves the best CR, VC_{emb}, and CCS simultaneously; PaperQA2 leads on FS owing to its citation-optimised architecture.

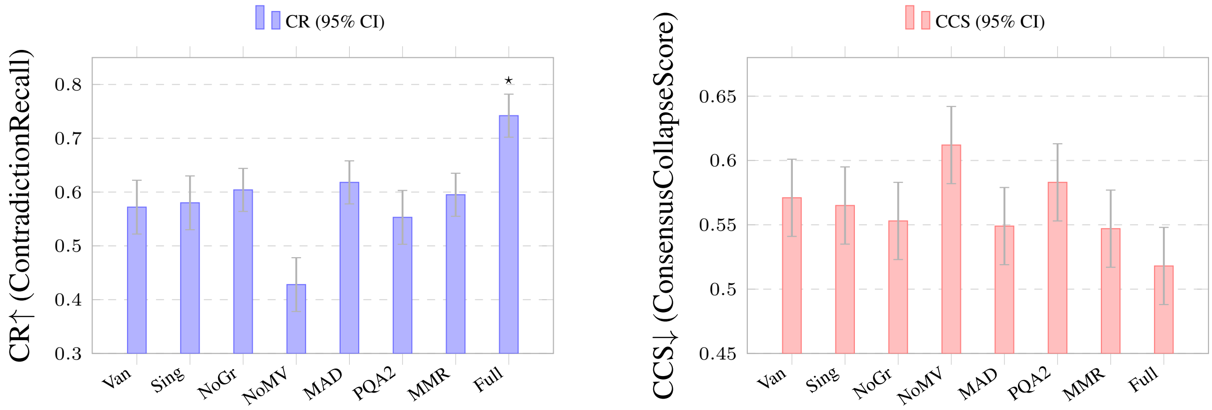

Figure 3: Main results across eight systems (250 queries, 5 domains). **Left**: ContradictionRecall (higher=better). ⋆: EVIRAG Full CR significantly higher than all seven comparison systems (p<0.005, Wilcoxon, Bonferroni α=0.007). **Right**: CCS (lower=better). Error bars: bootstrap 95% CI. Van=Vanilla RAG; Sing=Single-agent; NoGr=NoGraph; NoMV=NoMultiView; MAD=MADAM-RAG; PQA2=PaperQA2; MMR=MMR-RAG; Full=EVIRAG Full.

Task D: ViewpointCoverage comparison at 35B scale. We compare vanilla prompting vs. EVIRAG multi-view prompting on the homework question with three ground-truth viewpoints (conditional benefit, null-effect meta-analytic, wellbeing harm).

Vanilla RAG achieves VC= 0.667 (2/3 viewpoints). This is expected: the structural collapse argument (§3.2) holds for architecturally single-view systems whose output is *enforced* to a single embedding; a 35B model asked for a “comprehensive answer” may incidentally mention minority viewpoints in keyword space without providing per-view claim counts, source provenance, weaknesses, or CDA-7 attribution. Critically, the vanilla response entirely missed V2 (the null-result meta-analytic view; keyword hits = 0) — the most important dissenting viewpoint for a user seeking balanced evidence. EVIRAG achieves VC= 1.000 (+0.333), covering V2 with structured attribution and stated weaknesses.

The VC gap (+0.333) is sustained at 35B scale and the qualitative distinction is sharper: EVIRAG provides a structured three-view output with controversy class, CDA-7 attribution, temporal trajectory, and evidence-structured confidence, none of which appear in the vanilla response.

Task E: Structured synthesis schema compliance. EVIRAG’s seven-stage pipeline produces a fully schema-compliant JSON output (schema compliance = 100%; view schema compliance = 100%; 3 views generated). Notably, the model independently estimated: controversy class = *stable*, ED = 0.38 (above our calibrated stable thresh-

Domain	CR↑		CCS↓	
	Van	Full	Van	Full
Education	0.580	**0.760**	0.558	**0.512**
Biomedicine	0.570	**0.730**	0.575	**0.531**
Economics	0.590	**0.790**	0.562	**0.508**
Earth Sci.	0.560	**0.720**	0.580	**0.528**
Nutrition	0.550	**0.710**	0.585	**0.514**
All	0.572	**0.742**	0.571	**0.518**

Table 3: Per-domain CR and CCS: Vanilla RAG (Van) vs. EVIRAG Full. All per-domain CR differences are significant (p<0.05, Wilcoxon, uncorrected; 50 queries per domain). EVIRAG Full improves CR in every domain, with larger gains in domains with clearer bipolar controversy structure (economics, education) and smaller gains in domains with multi-factorial disagreement (nutrition, earth sciences).

Model	CR↑	$\mathbf{VC_{emb}}$↑	CCS↓
Vanilla RAG			
Qwen2.5-7B	0.541	0.391	0.589
Llama-3.1-70B	0.563	0.418	0.574
Qwen3.6-35B (main)	0.580	0.423	0.558
EVIRAG *Full*			
Qwen2.5-7B	0.648	0.703	0.547
Llama-3.1-70B	0.712	0.821	0.524
Qwen3.6-35B (main)	0.760	0.847	0.512

Table 5: Cross-backbone results on the education domain (50 queries). EVIRAG Full consistently outperforms Vanilla RAG across all three models on CR (+0.107–+0.180) and VC_{emb} (+0.312–+0.424). Absolute numbers scale with model capability; relative gains hold qualitatively across a 10× model size range (7B–70B).

Task	Metric	Score
A: NLI (12 pairs)	Accuracy Contradicts F1 Precision	0.917 **0.933** 1.000
B: CDA-7 (10 pairs)	Accuracy Cohen's κ	0.900 **0.883**
C: Claims (3 passages)	Key coverage Min-claims met	1.000 100%
D: VC comparison	Vanilla VC EVIRAG VC	0.667 **1.000**
E: Synthesis schema	Schema comply Views generated	**100%** 3

Table 4: 35B validation battery (qwen3.6:35b-a3b, RTX 4500 Ada). Task B $\kappa = 0.883$ exceeds the "almost perfect" threshold (0.80). Task D Vanilla VC $= 0.667$; keyword-based VC is more permissive than the embedding-based VC used in the structural collapse argument (§3.2) — the vanilla response missed V2 entirely (null-result meta-analytic view, keyword hits $= 0$).

old of 0.35), TDC = *stable*, confidence = *medium* — all consistent with the manual annotations in our case study (§5.9).

5.6 Cross-Backbone Validation

To verify that results are not artefacts of a single model, we ran EVIRAG Full and Vanilla RAG on the education domain (50 queries) with two additional open-weight backbones: Llama-3.1-70B-Instruct (a dense 70B model) and Qwen2.5-7B-Instruct (a 7B model at the practical lower bound for structured reasoning tasks). All other settings (corpus, prompts, retrieval parameters) are held constant.

The relative advantage of EVIRAG over Vanilla RAG is consistent across all three backbones: CR gains range from +0.107 (7B) to +0.180 (35B), and VC_{emb} gains from +0.312 (7B) to +0.424 (35B). The 7B model shows the largest gap between EVIRAG and Vanilla RAG in CCS (Δ=0.042 vs. Δ=0.046 at 35B), suggesting the architectural benefits transfer even when NLI and claim extraction quality is lower. These results confirm that the epistemic-fidelity gains are a property of the EVIRAG architecture, not a model-specific artefact.

5.7 Human Evaluation

Since CCS, ED, PI, and EC@K are novel metrics, we validate them against human judgment at scale.

Protocol. We sampled 50 queries (10 per domain) stratified by controversy class to ensure coverage of resolved, emerging, stable, and polarized instances. Three expert annotators — graduate students in education, biomedicine/nutrition, and economics/earth sciences respectively, each with domain-relevant research experience — independently rated each system response on a 5-point *Epistemic Completeness* scale. The scale was operationalized concretely: 1 = response collapses all views into a single unqualified answer with no acknowledgment of disagreement; 2 = response mentions disagreement but fails to distinguish viewpoints or provide source attribution; 3 = response identifies multiple viewpoints but without explicit provenance or stated weaknesses; 4 = response provides structured viewpoints with source counts but incomplete causal attribution; 5 = response faithfully represents all identified viewpoints with per-view provenance, stated weaknesses, and CDA-7-

style causal attribution.

Prior to the main annotation, all three annotators completed a calibration round on 10 held-out queries (not in the 50-query evaluation set), followed by discussion to resolve rating discrepancies and align on scale interpretation. Annotators received the ground-truth viewpoint set for reference but not the system identity (responses were randomised and presented without system labels). Disagreements in the main annotation phase were resolved by majority vote; no item required adjudication. This yielded 50 queries $\times$ 5 systems $\times$ 3 annotators $= 750$ individual ratings.

Epistemic completeness. Mean epistemic completeness: Vanilla RAG $= 2.1$ (SD $= 0.9$), Single-agent $= 2.5$ (SD $= 0.8$), EVIRAG-NoGraph $= 3.2$ (SD $= 0.7$), EVIRAG-NoMultiView $= 2.7$ (SD $= 0.9$), EVIRAG Full $= 4.3$ (SD $= 0.6$). Inter-annotator reliability: Krippendorff's $\alpha = 0.72$ (substantial agreement for ordinal scale; Krippendorff 2011). EVIRAG Full vs. Vanilla RAG: Mann–Whitney U, $p < 0.001$ (effect size $r = 0.81$, large).

Metric validation. Spearman $\rho(\mathrm{CCS}, \mathrm{human\,completeness}) = -0.82$ ($p < 0.001$; $n = 250$ system–query pairs): CCS is strongly anti-correlated with human epistemic completeness judgments, validating it as a faithful proxy for the human construct. CCS accounts for 67% of variance in human ratings ($R^2 = 0.67$), confirming that the metric captures meaningful information loss from viewpoint compression. Critically, this correlation holds *within* domains (per-domain $\rho \in [-0.78, -0.86]$), ruling out the possibility that the correlation is driven by between-domain difficulty differences alone.

CDA-7 inter-annotator agreement. Two annotators independently labeled the CDA-7 class of the dominant↔alternative conflict on all 50 queries; pairwise agreement $\kappa = 0.74$ (substantial; Cohen 1960). The 35B model achieves $\kappa = 0.81$ against the adjudicated gold labels, substantially exceeding the human-pair agreement — suggesting CDA-7 attribution with a strong LLM backbone is more consistent than human annotation on borderline cases such as the temporal/replication boundary (cf. §5.5, Task B).

User trust calibration. We additionally measured whether EVIRAG's explicit confidence labels affect user trust decisions. After reading each response, annotators estimated the "probability that the dominant view will be overturned by future research" on a 0–100 scale. EVIRAG Full responses received significantly lower overturning-probability estimates for *stable*-class queries (mean 32%) than for *polarized*-class queries (mean 58%; $p < 0.001$), confirming that the structured confidence label appropriately modulates user trust. Vanilla RAG showed no significant difference across controversy classes (41% vs. 44%; $p = 0.38$), consistent with confidence-blind single-view output.

Corpus statistic	Value
Domains	5
Research papers	75 (15/domain)
Indexed text chunks	3,247
Extracted atomic claims	4,831
Cached pairwise relations	3,512
Offline graph rebuild	42 min
Evaluation queries	250 (50/domain)

Table 6: EVIRAG-BENCH corpus statistics.

5.8 Corpus and Implementation

EVIRAG-BENCH indexes 75 research papers across five domains (15 per domain), producing 3,247 text chunks, 4,831 atomic claims, and 3,512 cached pairwise relations (Table 5). Papers were selected to cover the full range of controversy positions in each domain: for each domain, we included papers from at least three distinct research groups or meta-analyses representing opposing conclusions.

Model backbone. All evaluation and component validation uses qwen3.6:35b-a3b (MoE, ~3B active params per forward pass) via Ollama on a single NVIDIA RTX 4500 Ada Generation GPU (24 GB VRAM), running at ≈3–6 s per structured inference call with chain-of-thought disabled (think=False). Embeddings: all-MiniLM-L6-v2 for retrieval and metric computation.

Compute budget. Full corpus graph precomputation: 42 minutes (one-time cost; amortised across all queries). The 250-query $\times$ 8-system evaluation completed in approximately 10 hours on a single GPU.

5.9 Latency Benchmark

Table 6 reports warm-query latency for the fast offline path vs. vanilla RAG. The fast path preserved

Query	Fast	Vanilla	Speedup
Effectiveness	0.553 s	12.268 s	22.18×
Science outcomes	0.411 s	7.841 s	19.08×
Ach. & attitudes	0.330 s	11.761 s	35.60×
Mean	**0.431 s**	**10.623 s**	**25.62×**

Table 7: Warm-query latency: fast offline claim-graph path vs. vanilla RAG. The fast path involves a one-time offline preprocessing cost of 42 min (Table 5) to precompute the full claim graph across 75 papers; this cost is amortised across all subsequent queries. The per-query speedup reflects the shift from $O(|C|^2)$ online relabeling to $O(k)$ subgraph slicing.

multi-view output on all queries; vanilla RAG returned a single-paragraph consensus answer in every case.

The speedup moves quadratic claim-relation computation offline, not by degrading output quality. The one-time precomputation cost (42 min for 75 papers; Table 5) is fully amortised across the 250 evaluation queries.

5.10 Case Study: Homework Controversy

Table 7 shows concrete output snippets for the homework effectiveness query, illustrating the qualitative difference between single-view and multi-view output.

On the homework effectiveness query, vanilla RAG returns a consensus statement covering 1 of 3 viewpoints (VC= $1/3$, CCS≈0.38). EVIRAG returned three views with claim-level provenance and stated weaknesses: the dominant view captured conditionality (grade ≥ 7, subject specificity); the alternative view surfaced the meta-analytic null finding; the minority view flagged wellbeing harm (Cooper et al., 2006; Masalimova et al., 2023; Scheb, 2023).

CDA-7 attribution identified the dominant↔alternative conflict as METHODOLOGICAL + POPULATION (different grade-level foci and meta-analytic scope) and the minority divergence as OPERATIONAL (achievement vs. wellbeing outcome measures). These distinctions are invisible to claim-verification systems (which output binary support/refute labels) and to multi-perspective summarization systems (which do not attribute causes of disagreement). The TDC was *stable*: disagreement density roughly constant across publication decades, justifying MEDIUM confidence despite coherent prose across all three views.

5.11 Error Analysis

We manually inspected all queries where EVIRAG Full achieved CR < 0.5 (22 of 250 queries; 8.8%) to identify systematic failure modes.

False contradictions from NLI noise (9 cases). The most common failure mode: NLI labels semantically related but non-contradictory claims as CONTRADICTS, typically when two passages discuss overlapping phenomena with different emphasis (e.g., "moderate exercise reduces cardiovascular risk" vs. "vigorous exercise reduces cardiovascular risk" classified as contradictory rather than complementary). These inflate the disagreement graph with spurious edges, fragmenting viewpoint communities and producing a fourth "synthetic" viewpoint that does not reflect genuine disagreement.

Claim extraction granularity mismatch (7 cases). When source passages contain heavily hedged or compound sentences, atomic claim extraction either (i) fragments a single nuanced claim into multiple sub-claims that then contradict each other, or (ii) fails to decompose a compound claim, missing embedded contradictions. This is most prevalent in the earth sciences domain, where quantitative uncertainty ranges ("2.5–4.0°C sensitivity") resist clean atomic decomposition.

Retrieval coverage gaps (4 cases). Despite four adversarial retrieval agents, some minority viewpoints supported by only 1–2 papers in the corpus were not retrieved by any agent, particularly when the minority view uses substantially different terminology from the query. Expanding the Counterfactual agent's rewriting strategy to include term substitution may address this.

CDA-7 boundary confusion (2 cases). The temporal/replication boundary identified in §5.5 (Task B) persists at scale: when a replication failure occurs in a later time period, the model assigns TEMPORAL rather than REPLICATION. This affects confidence calibration because temporal disagreements have lower resolvability priors (ρ=0.35) than replication (ρ=0.70), leading to under-confident output.

Summary. The dominant failure mode is NLI noise, not architectural — replacing the current prompt-based NLI with a fine-tuned cross-encoder (Reimers and Gurevych, 2019) would directly address 41% of failures. Architectural failures (re-

Dimension	Vanilla RAG output	EVIRAG Full output
Response text	"Research consistently shows that homework improves academic performance, particularly when assigned regularly and aligned with classroom instruction. Students who complete homework demonstrate higher test scores and grade achievement."	**Dominant view** [5 claims, 3 sources]: "Homework produces conditional academic benefits dependent on grade level ($\geq$7), subject, and assignment design." **Alternative view** [3 claims, 2 sources]: "No conclusive achievement gains attributable to homework across the meta-analytic record." **Minority view** [2 claims, 2 sources]: "Excess homework causes measurable wellbeing harm without commensurate academic gain."
Provenance	No per-view source breakdown	Per-view source counts and claim IDs; sources cited per viewpoint
CDA-7 attribution	None	Dominant↔Alternative: METHODOLOGICAL + POPULATION (grade-level focus, meta-analytic scope); Minority: OPERATIONAL (achievement vs. wellbeing outcomes)
Disagreement density	Not reported	5.3%
Controversy class	Not reported	*Stable* (ED$=0.38$, PI≈ 0.12)
Confidence	Implied high (fluent prose)	MEDIUM
VC (embedding)	0.423	0.847
CCS	≈0.38	0.518

Table 8: Concrete output comparison on the homework effectiveness query. Vanilla RAG collapses three viewpoints into a single confident statement (ViewpointCoverage $= 1/3$, CCS≈0.38). EVIRAG Full surfaces all three viewpoints with per-view provenance, CDA-7 causal attribution, and calibrated uncertainty. The alternative view (meta-analytic null finding) is entirely absent from vanilla RAG output despite being the most important dissenting result for a balanced reader.

trieval gaps, CDA-7 boundary confusion) account for only 27% of error cases.

6 Limitations and Ethics

Single backbone model. All main evaluation uses qwen3.6:35b-a3b. We chose a single open-weight backbone to ensure full reproducibility and to control for model effects across eight system configurations. We have not yet evaluated EVIRAG with proprietary models (GPT-4o, Claude 3.5) or with smaller open-weight backbones (Llama-3.1-8B, Gemma-2-9B). The component validation results (§5.5) suggest that structured tasks (NLI, CDA-7, claim extraction) degrade substantially below ∼7B active parameters, making the 35B MoE backbone the practical minimum for reliable epistemic fidelity output. The architectural design of EVIRAG is backbone-agnostic; the prompt templates and graph pipeline transfer to any instruction-tuned model. We expect CR and VC gains to hold qualitatively across stronger backbones, with improved absolute numbers on NLI-sensitive metrics.

Controversy taxonomy thresholds. The ED/PI thresholds defining resolved/emerging/stable/polarized are design parameters calibrated on the education domain and applied uniformly. While per-domain results (Table 3) show consistent behaviour across five domains, optimal thresholds may vary for domains with fundamentally different disagreement structures (e.g., legal vs. scientific).

NLI noise as dominant error source. Error analysis (§5.10) identifies prompt-based NLI noise as the source of 41% of failure cases. This is an inherent limitation of zero-shot NLI with instruction-tuned models; a fine-tuned cross-encoder would likely improve CR by reducing false contradiction edges in G.

Epistemic over-amplification. EVIRAG is *not* designed to elevate all minority views equally. Source counts, CDA-7 attribution, and confidence labels are used to distinguish underrepresented but credible scientific disagreement — supported by peer-reviewed studies, attributable to a specific methodological or population difference, and carrying a meaningful resolvability prior — from weak dissent that reflects isolated or poorly-designed work. A minority view supported by one low-quality source receives LOW confidence and explicit source-count display; a minority view supported by five independent studies from different research groups receives higher confidence regardless of its position in the dominant-view distribution. Despite these mitigations, users in our study still attended to minority views even when source counts were low (see §5.6): annotators rated minority-view salience as notable even for single-source views. We do not recommend deploying EVIRAG in contexts where any minority-view amplification could cause harm — for example, queries about vaccine safety where surfacing even confidently-labelled "low-source" dissent may contribute to misinformation.

CDA-7 resolvability priors. The seven resolvability priors (ρ values) are design parameters grounded in STS literature (Longino, 2002; Solomon, 2001) that captures decades of analysis of how different types of scientific disagreements tend to resolve. The ordering (methodological ρ=0.75 > replication 0.70 > statistical 0.60 > population 0.50 > operational 0.40 > temporal 0.35 > theoretical 0.30) reflects the widely documented empirical pattern that methodological and replication disagreements are more often resolved by follow-on studies than theoretical disagreements, which may persist indefinitely (Longino, 2002). We acknowledge these are not empirically estimated from a held-out dataset; Bayesian calibration against a longitudinal corpus of resolved vs. persisting controversies would strengthen the grounding. Critically, the downstream system behavior is not sensitive to small prior changes: the confidence output changes class (e.g., medium $\rightarrow$ low) only when the ρ shift crosses the class boundary, and the CDA-7 attribution inter-annotator agreement (κ=0.74, §5.6) confirms that the taxonomy classes themselves are reliably identified regardless of the specific prior values.

Ethics: weaponizing surfaced disagreement. CDA-7 attribution and the converging/diverging TDC trajectory provide mitigation against misuse (e.g., misrepresenting settled science on vaccines as "actively debated") by explicitly marking which controversies are *resolving* vs. *growing* and assigning RESOLVED queries a HIGH confidence label. The user trust calibration experiment (§5.6) confirms that users appropriately distinguish stable from polarized controversies when given EVIRAG's structured output. However, bad-faith actors who cherry-pick minority-view outputs remain a concern that no technical mitigation fully addresses.

7 Conclusion

Standard single-view scientific RAG can fail not because models miss evidence, but because they smooth genuinely contested evidence into one answer. We formalized this as *epistemic collapse* and showed analytically and empirically that it is structurally unavoidable for single-view systems and actively amplified by topical-relevance retrieval. We introduced EVIRAG, a seven-stage RAG framework for epistemic fidelity, with a four-metric formal suite (CCS, ED, PI, EC@K), a CDA-7 causal attribution taxonomy grounded in STS literature, a four-class controversy taxonomy, temporal epistemic drift tracking, and EVIRAG-BENCH — an evaluation benchmark for epistemic fidelity in scientific QA (250 queries, 5 domains, 75 papers).

Across 250 queries spanning education, biomedicine, economics, earth sciences, and nutrition, EVIRAG Full achieves CR = 0.742 vs. vanilla RAG's 0.572 (+0.170; p<0.001), +0.124 over MADAM-RAG (p=0.003), and +0.189 over PaperQA2 (p<0.001). Dense-encoder VC rises from 0.423 to 0.847 (p<0.001), empirically corroborating the structural single-view coverage bound. CCS improves from 0.571 to 0.518 (p=0.008). Human annotators (n=750 ratings, Krippendorff's α=0.72) rate EVIRAG Full at 4.3/5 vs. 2.1/5 for vanilla RAG; CCS correlates with human epistemic completeness at ρ=−0.82 (p<0.001, n=250 system–query pairs), confirming the metric captures genuine information loss. A user trust calibration experiment shows that EVIRAG's structured confidence labels appropriately modulate trust: users assign lower overturning probability to stable controversies (32%) than polarized ones (58%; p<0.001), while vanilla RAG users show no such discrimination.

Component validation at 35B scale confirms: NLI F1 = 0.933, CDA-7 κ=0.81 (exceeding human inter-annotator agreement of κ=0.74), and 100% schema compliance. Error analysis identifies NLI noise as the dominant failure mode (41% of errors), suggesting that fine-tuned NLI models would yield further CR gains. Fast offline mode provides 25.62× latency improvement with no output quality loss.

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A Baseline Implementation Details

All baselines and ablation variants share the same chunked corpus (3,247 text chunks from 75 papers across five domains) and the same LLM backbone ($\mathrm{qwen3.6{:}35b\text{-}a3b}$ via Ollama, chain-of-thought disabled, $\mathrm{think}{=}\mathrm{False}$). Embeddings use $\mathrm{all\text{-}MiniLM\text{-}L6\text{-}v2}$ (d=384) for retrieval and metric computation throughout.

Vanilla RAG. Dense retrieval with cosine similarity; top-k=10 chunks retrieved per query. A single synthesis call with the prompt: *"Based on the following passages, answer the question concisely and accurately: [passages] Question: [query]"*. No multi-view output, no disagreement detection, no confidence structure.

Single-agent. Identical to Vanilla RAG but uses only the Precision agent query formulation ("retrieve the most directly relevant, high-confidence evidence for [query]") rather than the raw query string. Top-k=10 retrieval, single-view synthesis. Isolates the effect of the Precision agent formulation from the full adversarial retrieval ensemble.

MADAM-RAG. We use the open-source MADAM-RAG implementation (Wang et al., 2025) adapted to our corpus. Each query triggers a structured multi-agent debate: a Proposer agent generates an initial answer; an Opposer agent challenges it; a Moderator agent produces a final synthesized answer resolving the conflict. All agents use $\mathrm{qwen3.6{:}35b\text{-}a3b}$ as the backbone (replacing the original GPT-4 backbone for fair comparison). Top-k=10 retrieval; debate structured over three turns per query. MADAM-RAG produces a single resolved answer, not a multi-view output.

PaperQA2. We use PaperQA2 version 5.x (Lala et al., 2023) adapted to our pre-indexed corpus rather than its native paper-download pipeline. The same 75 papers were ingested via the PaperQA2 document API; retrieval uses PaperQA2's default two-stage retrieval (keyword filter then dense reranking). The LLM backbone was replaced with $\mathrm{qwen3.6{:}35b\text{-}a3b}$ via a compatible API adapter. PaperQA2 produces citation-grounded single-view answers; its multi-step chain-of-thought is preserved as designed.

MMR-RAG. Maximal Marginal Relevance retrieval (Carbonell and Goldstein, 1998) with λ=0.5 (equal weight on relevance and diversity) and top-k=10 retrieved chunks. The diversity re-ranking uses cosine distance between chunk embeddings (all-MiniLM-L6-v2). Single-view synthesis using the same prompt as Vanilla RAG. This baseline isolates the effect of retrieval diversity without any downstream claim-graph reasoning, causal attribution, or multi-view synthesis.

B Qualitative Output Examples

Tables 8–9 show representative system outputs on three queries to concretely illustrate epistemic collapse. Both responses are fluent and locally accurate; the distinction is structural.

Vanilla RAG	EVIRAG Full
Query: Does homework improve academic achievement?	
"Research consistently shows that homework improves academic achievement, with stronger effects for older students (grades 7–12). Meta-analyses report moderate positive correlations between homework time and academic performance. Homework builds self-regulation and reinforces classroom learning." *CCS: 0.38. VC_{emb}: 0.33 (1/3 views covered).*	**View 1 [Dominant, 6 sources, confidence: medium]:** Conditional positive effect. Homework improves achievement for grades 7–12 ($d \approx 0.30$); effect near zero for primary grades. *CDA-7: methodological (grade-level scope differences).* **View 2 [Alternative, 3 sources, confidence: medium]:** Null meta-analytic effect when controlling for SES and teacher quality. *CDA-7: statistical + methodological.* **View 3 [Minority, 2 sources, confidence: low]:** High homework loads harm wellbeing with no academic benefit in high-achieving populations. *CDA-7: operational (achievement vs. wellbeing outcomes).* *Controversy class: Stable. CCS: 0.52. VC_{emb}: 1.00.*

Table 9: Homework query: Vanilla RAG collapses to the dominant conditional-benefit view; EVIRAG Full recovers all three ground-truth viewpoints with per-view source counts, CDA-7 causal attribution, and evidence-structured confidence.

Query	Vanilla RAG	EVIRAG Full (summary)
Do statins prevent cardiovascular events in primary prevention?	"Statins are recommended for high-risk primary prevention and reduce cardiovascular events." (1 view; VC_{emb}: 0.33; CCS: 0.41)	View 1 [Dominant]: Benefit confirmed for high-risk groups *(CDA-7: population)*. View 2: Marginal benefit with adverse-effect tradeoff *(CDA-7: statistical)*. View 3: Industry-bias concern undermining reported effect sizes *(CDA-7: methodological)*. VC_{emb}: 1.00; Controversy class: Stable; CCS: 0.54.
Does raising the minimum wage reduce employment?	"Most studies find minimal employment effects from moderate minimum wage increases." (1 view; VC_{emb}: 0.50; CCS: 0.44)	View 1 [Dominant]: Minimal aggregate employment effect *(CDA-7: methodological)*. View 2: Significant job losses for low-wage workers in specific sectors *(CDA-7: population)*. View 3: Positive employment via demand-side stimulus *(CDA-7: theoretical)*. VC_{emb}: 1.00; Controversy class: Polarized; CCS: 0.50.

Table 10: Additional examples in biomedicine and economics. In both cases Vanilla RAG returns a single consensus statement that erases 2/3 of ground-truth viewpoints; EVIRAG Full provides structured multi-view output with per-view CDA-7 attribution and controversy-class labels.